Super-twisting lecture notes

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1 Introduction

1.1 By the end of this lecture

- You should have an overview of super-twisting sliding mode control, strengths, weaknesses, variations and solutions

1.2 Sliding mode recap

Second-order system

$$\begin{aligned}\dot{x}_1 &= x_2 \\ \dot{x}_2 &= h(x) + g(x)u\end{aligned}$$

where h and g are unknown nonlinear functions of x and $g(x) \geq g_0 > 0, \forall x$. If we want to stabilize the origin we can choose

$$s = \lambda x_1 + x_2$$

with derivative

$$\dot{s} = \lambda \dot{x}_1 + \dot{x}_2 = \lambda x_2 + h(x) + g(x)u$$

if $s = 0$ we obtain our sliding manifold (surface). This gives the dynamics

$$\dot{x}_1 = -\lambda x_1$$

Meaning that the dynamics go exponentially to zero on the sliding manifold.

Therefore, we want to choose a control law that sends the trajectory to the sliding manifold and stays there, i.e. $s = 0$ and $\dot{s} = 0$. Through Lyapunov analysis we end up with the function

$$u = -\beta(x) sgn(s)$$

where

$$sgn(s) = \begin{cases} 1, & s > 0 \\ 0, & s = 0 \\ -1, & s < 0 \end{cases}$$

The gain must be such that $\beta(x) \geq \varrho(x) + \beta_0, \beta_0 > 0$, for a value $\varrho(x)$ satisfying the inequality

$$\left| \frac{\lambda x_2 + h(x)}{g(x)} \right| \leq \varrho(x)$$

Remark 1. *The above statement means that we must have a control gain of a certain size in order to dominate the dynamics of the system. This implies that we need to have some knowledge of h and g in order to correctly tune the controller.*

2 Adaptive super-twisting algorithm

2.1 Intro

Key problems to solve

1. Chattering
2. Control gain uncertainty and overestimation
3. Input gain uncertainty

2.1.1 System

Given a single-input uncertain nonlinear system on the form

$$\dot{x} = f(x,t) + h(x,t)u \tag{1}$$

where $x \in \mathbb{R}^n$ is the state vector, $u \in \mathbb{R}$ is a control function, $f(x,t) \in \mathbb{R}$ is a differentiable, partially know vector-field.

2.1.2 Assumptions

Assumption 1. *A sliding variable $\sigma = \sigma(x,t) \in \mathbb{R}$ is designed so that the system's (1) desirable compensated dynamics are achieved in the sliding mode $\sigma = \sigma(x,t) = 0$.*

Assumption 2. *The system's (1) input-output ($u \to \sigma$) dynamics are of a relative degree one, and the internal dynamics are stable. Therefore, the input-output dynamics can be presented*

$$\begin{aligned} \dot{\sigma} &= \underbrace{\frac{\partial \sigma}{\partial t} + \frac{\partial \sigma}{\partial x} f(x,t)}_{\phi(x,t)} + \underbrace{\frac{\partial \sigma}{\partial x} h(x) u}_{b(x,t)} \\ \dot{\sigma} &= \phi(x,t) + \omega, \quad \omega = b(x,t)u \iff u = b^{-1}(x,t)\omega \end{aligned} \tag{2}$$

The solution of system (2) is understood in the sense of Filippov [1].

Remark 2. *We write down*

$$\dot{\sigma} = \phi(x,t) + b(x,t)u$$

Remark 3. *Sense of Filippov, from Sliding Mode Control and Observation [2]*

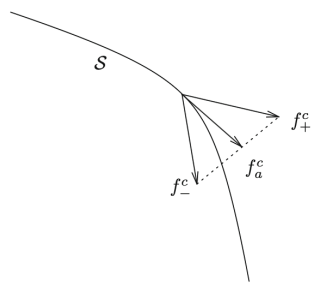
For a discontinuous, or an otherwise continuous system with discontinuous control, we can write the differential equation as:

$$\dot{x} = f^c(x)$$

where f^c is discontinuous with respect to x.

Main idea behind the solutions proposed by Filippov for differential equations with discontinuous right-hand sides is to construct a solution as the "average" of the solutions obtained by approaching the point of discontinuity from different angles.

The average solution is chosen to be tangent to the manifold $\mathcal{S}$ at the point of discontinuity.



Assumption 3. *Further, assume that function $b(x,t) \in \mathbb{R}$ is known and $b(x,t) \neq 0$ $\forall x$ and $t \in [0,\infty)$.*

Assumption 4. *the perturbation function $\phi(x,t) \in \mathbb{R}$ is bounded*

$$|\phi(x,t)| \leq \delta |\sigma|^{1/2} \tag{3}$$

where the finite boundary $\delta > 0$ exists but is not known.

Remark 4. *Means the perturbations are assumed to be vanishing with the square root of the sliding variable.*

2.2 The adaptive STW controller

2.2.1 Controller

The STW controller as written by Shtessel et al. [3] is given by

$$\begin{aligned} \omega &= -\alpha|\sigma|^{1/2} sgn(\sigma) + v \\ \dot{v} &= -\beta sgn(\sigma) \end{aligned} \tag{4}$$

where $\alpha = \alpha(\sigma, \dot{\sigma}, t)$ and $\beta = \beta(\sigma, \dot{\sigma}, t)$ are adaptive gains. The closed-loop sliding dynamics are then on the form

$$\begin{aligned} \dot{\sigma} &= -\alpha|\sigma|^{1/2} sgn(\sigma) + v + \phi(x,t) \\ \dot{v} &= -\beta sgn(\sigma) \end{aligned} \tag{5}$$

Remark 5. *Write down closed-loop dynamics*

2.2.2 Adaptive gains

Theorem 1 (Shtessel et al. [3])**.** *Consider system* (5)*. Suppose that the perturbation $\phi(x,t)$ satisfies* (3) *for some unknown constant $\delta > 0$. Then for any initial conditions $x(0), \sigma(0)$ the sliding surface $\sigma = 0$ will be reached in finite time via STW control* (4) *with the adaptive gains*

$$\begin{aligned} \dot{\alpha} &= \begin{cases} \omega_1 \sqrt{\frac{\gamma_1}{2}}, & \text{if } \sigma \neq 0, \\ 0, & \text{if } \sigma = 0 \end{cases} \\ \beta &= 2\epsilon\alpha + \lambda + 4\epsilon^2 \end{aligned} \tag{6}$$

where $\epsilon, \lambda, \gamma_1$ and ω_1 are arbitrary positive constants.

Remark 6. *I don't have a good answer to how they came up with this, but we can go through the proof together and see the justification for it*

2.2.3 Proof of Theorem 1

Proof in 2 steps

1. Change of variable
2. Lyapunov stability analysis to prove finite time convergence to the sliding manifold

Proof. To simplify the Lyapunov analysis we perform a change of variable

$$z = \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} = \begin{bmatrix} |\sigma|^{1/2} sgn(\sigma) \\ v \end{bmatrix}$$

then rewrite (5) to

$$\begin{cases} \dot{z}_1 &= \frac{1}{|z_1|}\left(-\frac{\alpha}{2} z_1 + \frac{1}{2} z_2 + \frac{1}{2}\phi(x,t)\right) \\ \dot{z}_2 &= -\frac{\beta}{|z_1|} z_1 \end{cases} \tag{7}$$

Equation (7) can then be written on matrix form

$$\dot{z} = \underbrace{\frac{1}{|z_1|}\begin{bmatrix} -\frac{\alpha}{2} & \frac{1}{2} \\ -\beta & 0 \end{bmatrix}}_{A(z_1)} \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} + \underbrace{\frac{1}{|z_1|}\begin{bmatrix} \frac{1}{2} \\ 0 \end{bmatrix}}_{g(z_1)} \phi(x,t) \tag{8}$$

Remark 7 (Observe)**.** *We observe the following*

1. *If $z_1, z_2 \to 0$ in finite time then $\sigma_1, \sigma_2 \to 0$ in finite time, and*
2. *$|z_1| = |\sigma|^{1/2}$ and $sgn(z_1) = sgn(\sigma)$*

Lyapunov analysis: In the second step of the proof, the stability analysis of the system given by the above equation is performed. We introduce the LFC

$$V(z,\alpha,\beta) = V_0 + \frac{1}{2\gamma_1}(\alpha-\alpha^*)^2 + \frac{1}{2\gamma_2}(\beta-\beta^*)^2 \tag{9}$$

where

$$V_0(z) = z^\top P z = \left(\lambda + 4\epsilon^2\right) z_1^2 + z_2^2 - 4\epsilon z_1 z_2 \tag{10}$$

$$P = \begin{bmatrix} \lambda + 4\epsilon^2 & -2\epsilon \\ -2\epsilon & 1 \end{bmatrix}, \ \forall \lambda, \epsilon > 0.$$

where $\alpha^*, \beta^* > 0$ are some constants.

The derivative of the Lyapunov function candidate is

$$\dot{V}(z,\alpha,\beta) = \dot{z}^\top P z + z^\top P \dot{z} + \frac{1}{\gamma_1}(\alpha-\alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta-\beta^*)\dot{\beta} \tag{11}$$

We can bound the first two terms

$$\dot{V}_0 = \dot{z}^\top P z + z^\top P \dot{z} \leq -\frac{1}{|z_1|} z^\top \tilde{Q} z \tag{12}$$

where the symmetric matrix $\tilde{Q}$ is given by

$$\tilde{Q} - 2\epsilon I = \begin{bmatrix} 2\lambda\alpha + 4\epsilon(2\epsilon\alpha - \beta) - 2\left(\lambda + 4\epsilon^2\right)\delta & * \\ \left(\beta - 2\epsilon\alpha - \lambda - 4\epsilon^2\right) + 2\epsilon\delta & 4\epsilon \end{bmatrix} \tag{13}$$

Remark 8. *The term $-2\epsilon I$ is likely there to simplify the right-hand side*

To ensure positive definite $\tilde{Q}$ we enforce

$$\beta = 2\epsilon\alpha + \lambda + 4\epsilon^2 \tag{14}$$

The matrix $\tilde{Q}$ is positive definite with a minimal eigenvalue $\lambda_{\min}(\tilde{Q}) \geq 2\epsilon$ if

$$\alpha > \frac{\epsilon\delta^2 + \left(\lambda + 4\epsilon^2\right)(2\epsilon + \delta) + \epsilon}{\lambda} \tag{15}$$

We then have that

$$\dot{V}_0 \leq -\frac{1}{|z_1|} z^\top \tilde{Q} z \leq -\frac{\epsilon}{|z_1|} z^\top z = -\frac{\epsilon}{|z_1|}||z||^2 \tag{16}$$

Remark 9. *I don't quite understand where ther $-\frac{\epsilon}{|z_1|} z^\top z$ term comes from*

We also have that

$$\lambda_{\min}(P)\,||z||^2 \leq \underbrace{z^\top P z}_{V_0(z)} \leq \lambda_{\max}(P)\,||z||^2 \tag{17}$$

where $||z||^2 = z_1^2 + z_2^2 = |\sigma| + z_2^2$ and

$$|z_1| = |\sigma|^{1/2} \leq ||z||^2 \leq \frac{V_0^{1/2}(z)}{\lambda_{\min}^{1/2}(P)} \tag{18}$$

Remark 10. $\lambda_{min}^{1/2}(P)||z||^2 \leq V_0^{1/2}(z)$*, is this true? Shouldn't it be* $\lambda_{min}^{1/2}(P)||z|| \leq V_0^{1/2}(z)$

then

$$\dot{V}_0(z) \leq -r V_0^{1/2}, \ r = \frac{2\epsilon\lambda_{\min}^{1/2}(P)}{\lambda_{\max}(P)} \tag{19}$$

Remark 11. *Jeg skjønner ikke helt hvor r kommer fra heller*

We also have that

$$
\begin{aligned}
\dot{V}(z,\alpha,\beta) &= \dot{z}^\top P z + z^\top P \dot{z} + \frac{1}{\gamma_1}(\alpha-\alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta-\beta^*)\dot{\beta} \\
&\leq -\frac{1}{|z_1|} z^\top \tilde{Q} z + \frac{1}{\gamma_1}(\alpha-\alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta-\beta^*)\dot{\beta} \\
&\leq -rV_0^{1/2} + \frac{1}{\gamma_1}(\alpha-\alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta-\beta^*)\dot{\beta} \\
&= -rV_0^{1/2} - \frac{\omega_1}{\sqrt{2\gamma_1}}|\alpha-\alpha^*|\dot{\alpha} - \frac{\omega_2}{\sqrt{2\gamma_2}}|\beta-\beta^*| \\
&+ \frac{1}{\gamma_1}(\alpha-\alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta-\beta^*)\dot{\beta} + \frac{\omega_1}{\sqrt{2\gamma_1}}|\alpha-\alpha^*| + \frac{\omega_2}{\sqrt{2\gamma_2}}|\beta-\beta^*|
\end{aligned}
\tag{20}
$$

Using the inequality

$$
\left(x^2+y^2+z^2\right)^{1/2} \leq |x|+|y|+|z| \tag{21}
$$

using this for the Lyapunov candidate function

$$
-rV_0^{1/2} - \frac{\omega_1}{\sqrt{2\gamma_1}}|\alpha-\alpha^*|\dot{\alpha} - \frac{\omega_2}{\sqrt{2\gamma_2}}|\beta-\beta^*| \leq -\eta V(z,\alpha,\beta)^{1/2} \tag{22}
$$

where $\eta = \min(r,\omega_1,\omega_2)$.

Using equation (22) we can rewrite the upper bound on $\dot{V}$

$$
\begin{aligned}
\dot{V}(z,\alpha,\beta) \leq -\eta V(z,\alpha,\beta)^{1/2} &+ \frac{1}{\gamma_1}(\alpha-\alpha^*)\dot{\alpha} + \frac{1}{\gamma_2}(\beta-\beta^*)\dot{\beta} \\
&+ \frac{\omega_1}{\sqrt{2\gamma_1}}|\alpha-\alpha^*| + \frac{\omega_2}{\sqrt{2\gamma_2}}|\beta-\beta^*|
\end{aligned}
\tag{23}
$$

Assuming the adaptation law (6) makes the adaptive gains $\alpha(t)$ and $\beta(t)$ bounded there exists positive constants α^* and β^* such that $\alpha(t)-\alpha^* > 0$ and $\beta(t)-\beta^* > 0$, $\forall t \geq 0$. Using this assumption we can reduce the upper bound expression to

$$
\dot{V}(z,\alpha,\beta) \leq -\eta V(z,\alpha,\beta)^{1/2} + \xi \tag{24}
$$

where

$$
\xi = -|\alpha-\alpha^*|\left(\frac{1}{\gamma_1}\dot{\alpha} - \frac{\omega_1}{\sqrt{2\gamma_1}}\right) - |\beta-\beta^*|\left(\frac{1}{\gamma_2}\dot{\beta} - \frac{\omega_2}{\sqrt{2\gamma_2}}\right) \tag{25}
$$

We want to ensure $\xi = 0$

$$
\dot{\alpha} = \omega_1\sqrt{\frac{\gamma_1}{2}} \tag{26}
$$

$$
\dot{\beta} = \omega_2\sqrt{\frac{\gamma_2}{2}} \tag{27}
$$

If we choose $\epsilon = \frac{\omega_2}{2\omega_1}\sqrt{\frac{\gamma_2}{\gamma_1}}$ and differentiating β as given by (6) we get

$$
\dot{\beta} = 2\epsilon\dot{\alpha} = 2\epsilon\omega_1\sqrt{\frac{\gamma_1}{2}} = \omega_2\sqrt{\frac{\gamma_2}{2}} \tag{28}
$$

Observe. For the controller to converge in finite time the inequality (15). This implies that $\alpha(t)$ must increase according to (26) until (15) is met in order to guarantee a positive definite $\tilde{Q}$ matrix.

> Also, as soon as the sliding variable σ and its derivative converges to zero it does not make sense to increase $\alpha(t)$ and $\beta(t)$ by making $\dot{\alpha} = 0$ as $\sigma = 0$. - Shtessel et al. [3]

Hence, we obtain the adaptation law (6). □ □

2.3 Propositions

Proposition 1. *The adaptive gains $\alpha(t)$ and $\beta(t)$ are bounded.*

Proof. A solution to the adaptive gain $\alpha(t)$ in (6)

$$\alpha = \begin{cases} \alpha(0) + \omega_1 \sqrt{\frac{\gamma_1}{2}} t, & 0 \leq t \leq t_r \\ \alpha(0) + \omega_1 \sqrt{\frac{\gamma_1}{2}} t_r, & t > t_r \end{cases} \tag{29}$$

Hence, $\alpha(t)$ and $\beta(t)$ are bounded. □ □

Proposition 2. *The finite time t_r can be estimated as*

$$t_r \leq \frac{2V^{1/2}(0)}{\eta} \tag{30}$$

where $\eta = min(r, \omega_1, \omega_2)$.

Proof. Since the right-hand side of the inequality (15) is bounded and $\alpha(t)$ is increasing linearly with respect to time according to the adaptation law (6), it is met in finite time. The above inequality is found by integrating (24). □ □

Key takeaways

- Deals well with matched uncertainties
- Chattering is attenuated by having the switching term behind an integrator
- Adaptive gain law is obtained and drives the system to the desired state in finite time
- The input gain is assumed known
- Controller gains are increasing
- Many design variables with tuning lack of intuition

3 Generalized super-twisting algorithm

3.1 Motivational example

3.1.1 Velocity-tracking problem

$$\left(1 + \cos^2(q)\right) \ddot{q} + g \sin(q) + b \left(\dot{q} + \arctan(\dot{q})\right) = u$$

where $q \in \mathbb{R}$ and $\dot{q} \in \mathbb{R}$ are state variables and $u \in \mathbb{R}$ is the torque control input

By defining $e_1 = \dot{q} - \dot{q}_d$, we get the following error dynamics

$$\dot{e}_1 = \gamma(q)u - \gamma(q)g\sin(q) - \gamma(q)b(e_1 + \dot{q}_d) + \arctan(e_1 + \dot{q}) - \ddot{q} \tag{31}$$

where $\gamma(q) = 1/(1 + \cos^2(q))$. Here, the time-varying function $\gamma(q(t))$ is an uncertain time-varying control coefficient. Let us remark that $\gamma(q(t))$ is bounded by $0 < 1/2 \leq \gamma(q(t)) \leq 1$, with a bounded time derivative.

Use standard STA

$$u_{sta} = -k_1 \lfloor e_1 \rceil^{1/2} + z$$
$$\dot{z} = -k_2 \lfloor e_1 \rceil^0$$

where $\lfloor \cdot \rceil^a = |\cdot|^a sgn(\cdot)$. The closed-loop dynamics then becomes

$$\dot{e}_1 = -k_1 \gamma(q) \lfloor e_1 \rceil^{1/2} + \gamma(q) \left[z - g\right] \tag{32}$$

3.1.2 Solution

$$\begin{aligned}\dot{e}_2 = &- k_2\lfloor e_1\rceil^0 + \dot{\varphi}\underbrace{-bc\gamma k_1\lfloor e_1\rceil^{1/2} - bc\gamma k_2\int_0^t \lfloor e_1(\tau)\rceil^0 d\tau}_{u(t)} \\ &+ bc\gamma\int_0^t \left[\dot{\varphi}(\tau) - b\left(k_1\lfloor e_1(\tau)\rceil^{1/2} - e_2(\tau)\right)\right] d\tau\end{aligned} \tag{33}$$

3.2 Generalized super-twisting algorithm

3.2.1 Control law

The generalized super-twisting algorithm [4] is given by

$$\begin{aligned} u &= -k_1\phi_1(\sigma) + z \\ \dot{z} &= -k_2\phi_2(\sigma) \\ \phi_1(\sigma) &= \lceil\sigma\rfloor^{1/2} + \beta\sigma \\ \phi_2(\sigma) &= \frac{1}{2}\lceil\sigma\rfloor^0 + \frac{3}{2}\beta\lceil\sigma\rfloor^{1/2} + \beta^2\sigma \end{aligned} \tag{34}$$

3.3 Finding conditions on gains

Cases

- General case
- State-dependent perturbations and know input gain
- Time-dependent unknown input gain and perturbation
- Closing the loop

3.3.1 General case

Scalar system

$$\dot{\sigma} = \gamma(x,t)u + \varphi(x,t) \tag{35}$$

Assumption 5. *The functions $\gamma(\sigma,t)$ and $\varphi(\sigma,t)$ are Lipschitz continuous functions with respect to t, and $\gamma(\sigma,t)$, $\varphi(\sigma,t) \in \mathcal{C}^1$ with respect to σ.*

Assumption 6. *The uncertain input gain function is bounded by*

$$0 < k_m \leq \gamma(x,t) \leq k_M \tag{36}$$

Assumption 7. *The perturbation term $\varphi(\sigma,t)$ can be split into two components*

$$\varphi(\sigma,t) = \varphi_1(\sigma,t) + \varphi_2(\sigma,t), \tag{37}$$

where is vanishing, i.e. $\varphi_1(0,t) = 0,\ \forall t \leq 0$, and is bounded by

$$|\varphi_1(\sigma,t)| \leq \alpha|\phi_1(\alpha)| \tag{38}$$

Assumption 8. *The total time derivative of $\frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}$ is*

$$\frac{d}{dt}\left(\frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) = \underbrace{\frac{1}{\gamma}\frac{\partial\varphi_2}{\partial t} - \frac{\varphi_2}{\gamma^2}\frac{\partial\gamma}{\partial t}}_{\delta_1(\sigma,t)} + \underbrace{\left(\frac{1}{\gamma}\frac{\partial\varphi_2}{\partial\sigma} - \frac{\varphi_2}{\gamma^2}\frac{\partial\gamma}{\partial\sigma}\right)}_{\delta_2(\sigma,t)}\dot{\sigma} \tag{39}$$

where $\delta_1(\sigma,t)$ and $\delta_2(\sigma,t)$ are bounded by positive constants

$$|\delta_1(\sigma,t)| \leq \bar{\delta}_1,\ |\delta_2(\sigma,t)| \leq \bar{\delta}_2 \tag{40}$$

Closed-loop system

$$\dot{\sigma} = -k_1\gamma(\sigma,t)\phi_1(\sigma) + \varphi_1(\sigma,t) + \gamma(\sigma,t)\left(z + \frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) \tag{41}$$

the system can be broken down

$$\begin{aligned} \dot{\sigma}_1 &= \gamma(\sigma,t)\left(-k_1\phi_1(\sigma_1) + \frac{\varphi_1(\sigma_1,t)}{\gamma(\sigma,t)} + \sigma_2\right) \\ \dot{\sigma}_2 &= -k_2\phi_2(\sigma_1) + \frac{d}{dt}\left(\frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) \end{aligned} \tag{42}$$

where $\sigma_1 = \sigma$ and $\sigma_2 = z + \frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}$

Theorem 2 (Castillo et al. [4]). *Suppose that $\gamma(\sigma,t)$ and $\varphi(x,t)$ of system (35) satisfy (36) and (40). Then, the states σ_1 and σ_2 converge to zero and z converges to $-\varphi(x,t)$, globally and in finite time, if GSTA gains k_1, k_2 and $\beta > 0$ are design as follows for any $\varepsilon > 0$*

$$k_1 > \frac{(1+\varepsilon)}{4\varepsilon k_m}\left(\frac{\alpha}{k_m}(k_M - k_m) + \frac{\bar{\delta}_2\varepsilon}{\beta}(k_M + k_m + 2)\right) + \frac{(1+\varepsilon)}{4\varepsilon k_m}\sqrt{\Lambda_{k_1}} + \frac{\alpha}{k_m} \tag{43}$$

$$\Lambda_{k_1} = \left(\frac{\alpha}{k_m}(k_M - k_m) + \frac{\bar{\delta}_2\varepsilon}{\beta}(k_M + k_m + 2)\right)^2 + 8\varepsilon(k_M - km)\left(\frac{\bar{\delta}_2}{\beta} + \bar{\delta}_1\right) \tag{44}$$

$$k_M\left(2\sqrt{\underline{h}}\epsilon\bar{c} - \frac{\bar{c}}{2}\sqrt{\Delta_{k_2}}\right)^2 + 2\bar{\delta}_1 < k_2 < k_M\left(2\sqrt{\underline{h}}\epsilon\bar{c} + \frac{\bar{c}}{2}\sqrt{\Delta_{k_2}}\right)^2 + 2\bar{\delta}_1 \tag{45}$$

where

$$\Delta_{k_1} = 16\underline{h}\epsilon - 8\left(\frac{(1+\epsilon)(k_M - k_m)}{\underline{k}_1 k_m}\right) \times \left(\frac{(1+\epsilon)}{\underline{k}_1 k_m}\left[\frac{\bar{\delta}_2\alpha}{\beta} + 2\bar{\delta}_1\right] + \frac{\bar{\delta}_2\epsilon}{\beta} + \frac{\alpha}{k_M}\right),$$

$$\underline{h} = 1 - \frac{\bar{\delta}_2}{\beta}\frac{k_m + \epsilon}{(k_1 k_m - \alpha)},\ \bar{c} = \frac{\underline{k}_1 k_m}{(1+\epsilon)(k_M - k_m)},\ \underline{k}_1 = k_1 - \frac{\alpha}{k_m}$$

Moreover, a system trajectory starting at $\sigma(0) = (\sigma_1(0), |\sigma_2(0))$ reaches the origin in a time smaller than

$$T(\sigma(0)) = \frac{2}{\mu_2}\ln\left(\frac{\mu_2}{\mu_1}V^{1/2}(x_1(0), x_2(0)) + 1\right) \tag{46}$$

Choosing gains

3.4 State-dependent perturbations and know input gain

System

$$\dot{\sigma} = u + \varphi(x,t) \tag{47}$$

Assumption 9. *The function $\varphi(\sigma,t)$ is Lipschitz continuous functions with respect to t, and $\varphi(\sigma,t) \in \mathcal{C}^1$ with respect to σ.*

Assumption 10. *The total time derivative of $\frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}$ is*

$$\frac{d}{dt}\left(\varphi_2(\sigma,t)\right) = \underbrace{\frac{\partial\varphi_2}{\partial t}}_{\delta_1(\sigma,t)} + \underbrace{\left(\frac{\partial\varphi_2}{\partial\sigma}\right)}_{\delta_2(\sigma,t)}\dot{\sigma} \tag{48}$$

Closed-loop system

$$\dot{\sigma} = -k_1\phi_1(\sigma) + \varphi_1(\sigma,t) + \left(z + \frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) \tag{49}$$

the system can be broken down

$$\begin{aligned} \dot{\sigma}_1 &= \left(-k_1\phi_1(\sigma_1) + \varphi_1(\sigma_1,t) + \sigma_2\right) \\ \dot{\sigma}_2 &= -k_2\phi_2(\sigma_1) + \frac{d}{dt}\left(\varphi_2(\sigma,t)\right) \end{aligned} \tag{50}$$

where $\sigma_1 = \sigma$ and $\sigma_2 = z + \varphi_2(\sigma,t)$

Corollary 1 (1, Castillo et al. [4]). *Suppose that $\varphi(\sigma,t)$ of system (47) $\delta_1(\sigma,t)$ and $\delta_2(\sigma,t)$ are bounded by $\bar{\delta}_1$ and $\bar{\delta}_2$, respectively. Then, the states σ_1 and σ_2 converge to zero and z converges to $-\varphi(\sigma,t)$, globally in finite-time, if GSTA gains k_1, k_2 and $\beta > 0$ are designed as follows for any $\epsilon > 0$*

$$k_1 > \frac{2(1+\epsilon)\bar{\delta}_2}{\beta} + \alpha \tag{51}$$

$$k_2 > \frac{1}{4\underline{h}\epsilon}\left(\frac{(1+\epsilon)}{(k_1-\alpha)}\left(\frac{\bar{\delta}_2\alpha}{\beta} + 2\bar{\delta}_1\right) + \frac{\bar{\delta}_2\epsilon}{\beta} + \alpha\right)^2 \tag{52}$$

3.5 Time-dependent unknown input gain and perturbation

Scalar system

$$\dot{\sigma} = \gamma(t)u + \varphi(t) \tag{53}$$

Assumption 11. *The functions $\gamma(t)$ and $\varphi(t)$ are Lipschitz continuous functions with respect to t.*

Assumption 12. *The uncertain input gain function is bounded by*

$$0 < k_m \leq \gamma(t) \leq k_M \tag{54}$$

Assumption 13. *The total time derivative of $\frac{\varphi_2(t)}{\gamma(t)}$ is*

$$\delta_1(t) := \frac{d}{dt}\left(\frac{\varphi_2(t)}{\gamma(t)}\right) = \frac{1}{\gamma}\frac{\partial\varphi_2}{\partial t} - \frac{\varphi_2}{\gamma^2}\frac{\partial\gamma}{\partial t} \tag{55}$$

where $\delta_1(t)$ is bounded by positive a constant

$$|\delta_1(t)| \leq \bar{\delta}_1 \tag{56}$$

3.6 Time-dependent unknown input gain and perturbation

Closed-loop system

$$\dot{\sigma} = -k_1\gamma(t)\phi_1(\sigma) + \varphi_1(t) + \gamma(t)\left(z + \frac{\varphi_2(t)}{\gamma(t)}\right) \tag{57}$$

the system can be broken down

$$\begin{aligned} \dot{\sigma}_1 &= \gamma(t)\left(-k_1\phi_1(\sigma_1) + \frac{\varphi_1(t)}{\gamma(t)} + \sigma_2\right) \\ \dot{\sigma}_2 &= -k_2\phi_2(\sigma_1) + \frac{d}{dt}\left(\frac{\varphi_2(t)}{\gamma(t)}\right) \end{aligned} \tag{58}$$

where $\sigma_1 = \sigma$ and $\sigma_2 = z + \frac{\varphi_2(t)}{\gamma(t)}$

Corollary 2 (2, Castillo et al. [4]). *Suppose that $\gamma(t)$ and $\varphi(t)$ of system (53) satisfy (54) and (56). Then, the states σ_1 and σ_2 converge to zero and z converges to $-\frac{\varphi(t)}{\gamma(t)}$, globally and in finite-time, if GSTA gains k_1, k_2 and $\beta \geq 0$ are designed as follows for all $\epsilon > 0$*

$$k_1 > \frac{(1+\varepsilon)}{4\varepsilon k_m}\left(\frac{\alpha}{k_m}(k_M - k_m)\right) + \frac{(1+\varepsilon)}{4\varepsilon k_m}\sqrt{\Lambda_{k_1}} + \frac{\alpha}{k_m} \tag{59}$$

$$\Lambda_{k_1} = \left(\frac{\alpha}{k_m}(k_M - k_m)\right)^2 + 16\varepsilon(k_M - km)\bar{\delta}_1 \tag{60}$$

$$k_M\left(2\sqrt{\epsilon}\bar{c} - \frac{\bar{c}}{2}\sqrt{\Delta_{k_2}}\right)^2 + 2\bar{\delta}_1 < k_2 < k_M\left(2\sqrt{\epsilon}\bar{c} + \frac{\bar{c}}{2}\sqrt{\Delta_{k_2}}\right)^2 + 2\bar{\delta}_1 \tag{61}$$

where

$$\Delta_{k_1} = 16\epsilon - 8\left(\frac{(1+\epsilon)(k_M - k_m)}{\underline{k}_1 k_m}\right) \times \left(\frac{(1+\epsilon)}{\underline{k}_1 k_m}2\bar{\delta}_1 + \frac{\alpha}{k_M}\right),$$

$$\bar{c} = \frac{\underline{k}_1 k_m}{(1+\epsilon)(k_M - k_m)},\ \underline{k}_1 = k_1 - \frac{\alpha}{k_m}$$

3.7 Closing the loop

Example revisited

$$\left(1+\cos^2(q)\right)\ddot{q} + g\sin(q) + b\left(\dot{q} + \arctan(\dot{q})\right) = u$$

3.8 Key takeaways

Features

- Deals well with matched uncertainties
- Works for uncertain input gain
- Rigorously proves control gain conditions for desired stability properties

Future work

- Combine with adaptation law

4 Adaptive generalized super-twisting algorithm

4.1 Control law

Like in [4] the GSTA algorithm is given by

$$\begin{aligned} u &= -k_1\phi_1(\sigma) + z \\ \dot{z} &= -k_2\phi_2(\sigma) \\ \phi_1(\sigma) &= \lceil\sigma\rfloor^{1/2} + \beta\sigma \\ \phi_2(\sigma) &= \frac{1}{2}\lceil\sigma\rfloor^{0} + \frac{3}{2}\beta\lceil\sigma\rfloor^{1/2} + \beta^2\sigma \end{aligned} \tag{62}$$

4.1.1 Adaptive gain law

Using same adaptive law as [3]

$$\begin{aligned} \dot{k}_1 &= \begin{cases} \omega_1\sqrt{\frac{\gamma_1}{2}}, & \text{if } \sigma \neq 0, \\ 0, & \text{if } \sigma = 0 \end{cases} \\ k_2 &= 2\epsilon k_1 + \lambda + 4\epsilon^2 \end{aligned} \tag{63}$$

4.1.2 Closed-loop system

$$\dot{\sigma} = -k_1\gamma(\sigma,t)\phi_1(\sigma) + \varphi_1(\sigma,t) + \gamma(\sigma,t)\left(z + \frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) \tag{64}$$

the system can be broken down

$$\begin{aligned} \dot{\sigma}_1 &= \gamma(\sigma,t)\left(-k_1\phi_1(\sigma_1) + \frac{\varphi_1(\sigma_1,t)}{\gamma(\sigma,t)} + \sigma_2\right) \\ \dot{\sigma}_2 &= -k_2\phi_2(\sigma_1) + \frac{d}{dt}\left(\frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}\right) \end{aligned} \tag{65}$$

where $\sigma_1 = \sigma$ and $\sigma_2 = z + \frac{\varphi_2(\sigma,t)}{\gamma(\sigma,t)}$

Theorem 3 (Borlaug et al. [5])**.** *Suppose that $\gamma(\sigma_1,t)$ and $\varphi(\sigma_1,t)$ in system (35) satisfy Assumptions 1-4, from [4]. Then, the closed-loop dynamics (65) are globally finite-time stable, such that the states σ_1 and σ_2 converge to zero and z converges to $-\varphi(0,t)/\gamma(0,t)$, globally and in finite time, ig the gains k_1 and k_2 are designed as expressed in (63), $\beta > 0, \lambda > 0, \omega_1 > 0, \gamma_1 > 0$, and $\epsilon = \frac{\omega_2}{2\omega_1}\sqrt{\frac{\gamma_2}{\gamma_1}}$, where $\omega_2 > 0$ and $\gamma_2 > 0$.*

Remark 12.

- *The theorem and proof can be extended to n dimensions by separating the problem into n one-dimensional cases*
- *The proof is similar to that of Shtessel et al. [3]*

4.2 Real system

Equations of motion

$$\begin{aligned} \dot{\xi} = J(\eta_2)\zeta = \begin{bmatrix} R_{Ib}^\top(\eta_2) & 0_{3\times 3} & 0_{3\times(n-1)} \\ 0_{3\times 3} & J_{k,o}^{-1}(\eta_2) & 0_{3\times(n-1)} \\ 0_{(n-1)\times 3} & 0_{(n-1)\times 3} & I_{(n-1)\times(n-1)} \end{bmatrix} \zeta \\ M(q)\dot{\zeta} + C(q,\zeta)\zeta + D(q,\zeta)\zeta + g(q, R_{Ib}) = \tau(q) \end{aligned} \tag{66}$$

where $\xi = \left[\eta_1^\top, \eta_2^\top, q^\top\right]^\top$, here $\eta_1 = [x, y, z]^\top \in \mathbb{R}^{6+(n-1)}$ is the position, $\eta_2 = [\phi, \theta, \psi]^\top \in \mathbb{R}^3$ are Euler angles and $q \in \mathbb{R}^{n-1}$ are the joint angles. The velocity vector $\zeta = \left[\nu^\top, \omega^\top, \dot{q}^\top\right]^\top \in \mathbb{R}^{6+(n-1)}$ comprises body-fixed linear and angular vehicle velocities ν and ω, and joint velocities $\dot{q}$.

4.2.1 Tracking control law using AGSTA

We choose sliding manifold

$$\sigma = \zeta - \zeta \in R^{6+(n-1)} \tag{67}$$

and the control input is given by

$$\tau(q) = u_{\text{AGSTA}} \in R^{6+(n-1)} \tag{68}$$

4.3 Experiments

Tracking performance [5] Gain evolution [5]

4.4 Comparison

Performance [5]
Actuation [5]

4.5 Key takeaways

Features

- Deals well with matched uncertainties
- Works for uncertain input gain
- Practically proven control method with desirable stability guarantees
- AGSTA demonstrates better tracking performance with GSTA with constant gains

Future work

- Coupled MIMO formulation
- Adaptive law with fewer parameters
- Adaptive law that enables decreasing controller gain

5 Multivariable generalized super-twisting algorithm

5.1 Problem statement

5.1.1 Multivariable system

Consider the multivariable system

$$\dot{x} = f(t, x) + G(t, x)u \tag{69}$$

where $x \in \mathbb{R}^n$ is the state vector, $u \in \mathbb{R}^n$ is the contorl input, $f : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^n$ is an uncertain vector field containing uncertainties and/or perturbations, and $G : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^{n\times n}$ is the uncertain input matrix.

5.1.2 Input matrix

The uncertain input matrix can be rewritten

$$G(t,x) = (I + \Delta_G(t,x))\, G_0(t,x) \tag{70}$$

where $G_0 : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^{n\times n}$ corresponds to the nominal part, which is assumed to be invertible, and $\Delta_G : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^{n\times n}$ is the uncertain term

5.2 Multivariable super-twisting algorithm

5.2.1 Control law by Moreno et al. [6]

The MGSTA is given by

$$\begin{aligned} u &= -k_1\phi_1(x) + bG_0^{-1}(t,x)v \\ \dot{v} &= -k_2\phi_2(x) \end{aligned} \tag{71}$$

The nonlinear functions $\phi_i : \mathbb{R}^n \to \mathbb{R}^n$ are monotonically increasing, ϕ_1 is continuous everywhere and ϕ_2 is continuous everywhere except possibly at $x = 0$.

Nonlinear functions

$$\phi_1 = \left(\alpha||x||^{-p} + \beta\right) x \tag{72}$$

$$\phi_2 = J(x)\phi_1(x) = c(x)\phi_1(x) \tag{73}$$

5.2.2 Decomposed perturbations

The uncertain term $f(t,x)$ is assumed to be decomposed as

$$f(t,x) = f_1(t,x) + f_2(t,x) = \Delta_1(t,x)\phi_1(x) + f_2(t,x) \tag{74}$$

where the function $f_2(t,x)$ is such that

$$\frac{d}{dt}\left[\left(b\,(I+\Delta_G)^{-1} f_2\right)\right] = \Delta_2(t,x)\phi_2 + \Delta_3(t,x)\dot{x} \tag{75}$$

where $\Delta_i : \mathbb{R}_+ \times \mathbb{R}^n \to \mathbb{R}^{n\times n}, i = 1,2,3.$

For the dynamical system (69) and controller (71), (72) and (73), the following conditions hold for all $t \geq 0, x \in \mathbb{R}^n$.

Assumption 14. *There exit constants $g_M > 0, \delta_g \geq 0$ and $g_m^\Delta > 0$ such that*

$$\begin{aligned} ||G(t,x)|| &\leq g_M \\ ||\Delta_G(t,x)|| &\leq \delta_g \\ Sym\{G(t,x)\} &\geq g_m^\Delta I \end{aligned} \tag{76}$$

Assumption 15. *Matrices $\Delta_i, i = 1,2,3$, in (74) and (75), are bounded, there exist $\delta_i \geq 0$ such that*

$$||\Delta_i(t,x)|| \leq \delta_i, i = 1,2,3.$$

Assumption 16. *There exist positive constants $0 < \gamma_1^\Delta \leq \gamma_2^\Delta$ such that*

$$0 < \gamma_1^\Delta I \leq Sym\{\mathcal{J}(x)\,(I+\Delta_G)\} \leq \gamma_2^\Delta I \tag{77}$$

As a consequence of these assumptions, there also exist constants $\gamma_3, \gamma_4, \gamma_5 > 0$, and $0 < \gamma_1^\Delta \leq \gamma_2^\Delta$ such that

$$\begin{aligned} Gram\{\mathcal{J}G\} &\leq \gamma_3 I \\ 2Sym\{\Delta_G\mathcal{J}G\} &\leq \gamma_4 I \\ Gram\left\{\Delta_G^\top\right\} &\leq \gamma_5 I \end{aligned} \tag{78}$$

Closed-loop dynamics The closed-loop dynamics including MGSTA is given by

$$\dot{x} = f_1 + (I + \Delta_G)(-G_0 k_1 \phi_1 + bz) \tag{79}$$

$$\dot{z} = -k_2\phi_2 + \Delta_2\phi_2 + \Delta_3\dot{x} \tag{80}$$

Theorem 4 (Moreno et al. [6]). *Select $\alpha > 0, \beta > 0, b > 0$ and $p \in \left(0, \frac{1}{2}\right]$ and let Assumption 1 be satisfied. Assume further that the inequality*

$$\gamma_1^{\Delta} g_m^{\Delta} > \gamma_4 + 2\sqrt{\gamma_3\gamma_5} \tag{81}$$

is satisfied. Then, for arbitrary values of $\delta_2, \delta_3, \delta_3$, there exist positive constant controller gains k_1, k_2 such that $\left[\sigma^\top, z^\top\right]^\top$ is globally robust finite-stable.

5.2.3 Algorithm 1

For proper selection of the control gains k_1 and k_2 consult the following algorithm

1. Select $\alpha > 0, \beta > 0, b > 0$ and $p \in \left(0, \frac{1}{2}\right]$. The parameter p should be selecting on the size of the perturbations.
2. Select $p_2 = p_2^*$ and $p_1 = p_1^*$ such that
$$\gamma_1^{\Delta} > p_2\mu_4 \tag{82}$$
and
$$\gamma_1^{\Delta} g_m^{\Delta} - \gamma_4 - \Gamma_0(p_1, p_2) > 0 \tag{83}$$
are satisfied for $p_2 = p_2^*$ and $p_1 \geq p_1^*$. For proper selection of the control gains k_1 and k_2 consult the following algorithm
3. Find a value $p_1^{\#}$ of p_1 such that $p_1 \geq p_1^{\#}$ and
$$\Gamma_1(p_1, p_2)p_1 > \Gamma_2(p_1, p_2) \tag{84}$$
is satisfied for every $p_1 \geq p_1^{\#}$. This is always feasible. For proper selection of the control gains k_1 and k_2 consult the following algorithm
   (a) Fix $p_2 = p_2^*, p_1 \geq p_1^{\#}$ and $p_1 p_2 > 1$, choose
   $$k_2 = b\frac{p_1}{p_2}, \tag{85}$$
   and
   $$\frac{-\alpha_1 - \bar{\alpha}}{2\alpha_2} < k_1 < \frac{-\alpha_1 + \bar{\alpha}}{2\alpha_2} \tag{86}$$
   with $\bar{\alpha} = \sqrt{\alpha_1^2 - 4\alpha_2\alpha_0}$

5.3 Example: Four-wheeled omidirectional robot

Sliding variables, [6]
Position and orientation, [6]
Armature voltages, [6]

5.4 Key takeaways

Features

- Deals well with matched uncertainties
- Works for uncertain input gain
- Coupled MIMO formulation

Future work

- Implement adaptive law
- Cleaner formulations and communication

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